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(1) Research Program/Project/Study Title: ACCEPTABILITY OF BANGUS-CLAM SISIG	
Proponent Name:	JANET V. LEDESMA
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	ROXAS CITY
		Province:	CAPIZ
		Region:	VI
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Food Processing Sector Restaurants Small and Medium Enterprise Academe		(8) Discipline/Field of Specialization CULINARY	

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The Philippines' national fish, Bangus or milkfish, is widely available due to its adaptability and being bred on fish farms. Its mild flavor pairs well with various cooking methods and spices. Clams, high in nutritional density, are bivalve mollusks found in cuisines worldwide. Sizzling sisig, a Kapampangan delicacy, traditionally made with pig face and ears, has evolved into various adaptations, including the use of fish, seafood, tofu, and other meats..

This study addresses the growing demand for healthier food options by focusing on the adaptation of the traditional Filipino dish, sisig. The researcher's choice of using bangus (milkfish) and clam as the main ingredients aims to reduce the fat content of the dish while ensuring it remains palatable. This shift not only aligns with the market trend towards healthier choices but also capitalizes on the abundance of these raw materials in the local environment. The campus availability of these ingredients enhances the feasibility and practicality of incorporating these locally sourced options into sisig recipes, aligning with the study's goal of providing healthier and sustainable alternatives

The versatility of bangus makes it an excellent choice, given its adaptability and widespread availability in fish farms throughout Capiz. The distinctive and mild flavor of bangus complements the overall taste profile of sisig, providing a healthier alternative to traditional meat options.

Incorporating clams, known for their high nutritional density, adds both flavor and nutritional value to the dish. Clams, being bivalve mollusks, contribute to the overall sustainability of the seafood industry and are widely recognized for their vitamins and minerals.

The sizzling sisig, traditionally made with grilled pig face and ears, undergoes a transformation in this study. The adaptation includes using flaked fish, such as bangus, and incorporating clams alongside other alternatives like seafood, tofu, chicken, and crispy lechon kawali. This variety not only caters to diverse dietary preferences but also ensures that the dish remains a popular and versatile option for both appetizers and dinner entrees.

The primary goal is to offer a healthier choice without compromising the rich and flavorful experience associated with sisig. By optimizing the abundance of bangus and clams in the local context, this research aims to contribute to a sustainable and health-conscious culinary landscape. The findings of this study could potentially influence the market by introducing a nutritious and delicious alternative to the traditional sisig, reflecting the evolving preferences of consumers towards healthier dining options.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The main objective of this study is to utilize Bangus and clam for sisig production.

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The specific objectives of the study included the following:

1. To determine sensory characteristics of the MPF bangus-clam sisig
2. To determine the most preferred formulation (MPF) of bangus-clam sisig.
3. To determine the acceptability of the MPF bangus-clam sisig among the consumers.
4. To determine the proximate analysis of the MPF bangus-clam sisig.
5. To determine the microbial quality of bangus-clam sisig.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

PH22018000610U1 A COMPOSITION FOR MUD CLAMS (ANODONTIA EDENTULA) SALAD (SISIG) WITH COCONUT MILK

There are plenty of sisig recipes, however, no research has been conducted to utilize the mud clams (*Anodonta edentula*) or Imbao as salad (Sisig). This utility model relates to a composition of making mud clams (*Anodonta edentula*) salad (sisig). The composition includes 200 g. steamed fresh imbao for three minutes, 3 T. vegetable oil to sauté the 2/3 cup small cubed ginger, 1/4 c. chopped garlic, 1/4 c. small cubed onion bulb, 3T diced green and red pepper, 1/3 c. cubed chili pepper, adding 3 pcs. crushed bay leaf, 1/8t. ground pepper 1 t. salt to improve the taste, pouring of the pint c. of coconut milk gata to the 15 sautéed ingredients, letting it simmer for 1 minute, transferring the cooked mud clams (*Anodonta edentula*) salad (Sisig) with coconut milk; and topping 2 stalks spring onion to garnish the salad (sisig).

PH22019001306U1 PROCESS OF PRODUCING MUSHROOM SISIG

This utility model relates to a process of producing mushroom sisig comprising the following steps: (a) coating the mushroom and vegetables with flour; (b) adding of seasonings to the coated mushroom and vegetables, (c) frying the seasoned mushroom and vegetables, (d) draining the fried mushroom and vegetables, (e) chopping the drained fried mushroom and vegetables, (f) sautéing the chopped mushroom and vegetables with herbs and spices, (g) seasoning the sautéed mushroom and vegetables with flavorings and souring agents, (h) cooling the cooked mushroom and vegetable sisig, (i) packing the cooled mushroom and vegetable sisig in aluminum foil pouch, (j) sealing the mushroom and vegetables sisig, (k) labeling the individual pack of mushroom and vegetables sisig, and (l) storing the packed and labeled mushroom and vegetables sisig in a cool, dry place.

PH22016000709U1 FORMULATION OF MILKFISH (CHANOS CHANOS) DUMPLING FILLING

This utility model discloses the formulation of a convenient food item; Bangus or Chanos. Chanos fish dumplings from a mixture of bangus fish meat, shrimp, pork, carrots, singkamas, onions, raisins, raw egg and seasoning. Bangus is the most popular seafood dish among Filipinos, very abundant in any locality and the main aquaculture product of the Philippines. Utilizing this commodity into another way of preparation and presentation is therefore a welcome move.

(12) Scientific Basis/Theoretical Framework (Include concepts. theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Materials and Methods

Materials.

The following are the materials used in the preparation of Bangus-Clam Sisig: Knife, Chopping Board, Preparation Bowl, Weighing Scale, Pot & pans, Frying Pan, Ladle.

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Raw materials

The following are the raw materials used in the preparation of Bangus-Clam Sisig: Bangus, Clam, Garlic, Chili, Onion, Liver spread, Soy Sauce, Black Pepper, Annatto Oil, Salt.

METHODS

Preparation of Bangus

Clean and debone the bangus
Seasoned the boneless bangus
Coat the boneless bangus in flour and cornstarch
Fry the boneless bangus
Let it cool
Flakes the boneless bangus
Set aside

Preparation of Clams

Depurate clams in water with salt for 3hrs.
Wash thoroughly in running water, until totally clean
Steam for 20mins
Let it cool
Remove from the shell
Remove the intestine of the clam
Chop the meat clam
Set aside

Cooking Procedure of Bangus-Clams Sisig

Heat the annatto oil over low fire
Suttee garlic until light brown
Caramelized onion
Add clams, stir until tender
Add bangus flakes
Add remaining ingredients; liver spread, seasoning, vinegar, and black pepper
Mix all and taste
When done, remove from the heat.
Cool completely before storing in vacuum pack

Sensory Evaluation of Bangus- Clam Sisig

Sensory evaluation consists of judging the quality of food by a panel of 20 subject experts, 20 faculties of Capsu, and 60 students of Capsu. The evaluation deals with analyzing and interpreting the qualities of bangus-clam sisig as they are perceived by the sense of sight, taste, touch, and hearing. The 9-point Hedonic scale score-card method was used to determine the sensory characteristics of bangus-clam sisig develop. Sensory attributes of the bangus-clam sisig namely, appearance, color, flavor, texture, taste, aroma, and consumer acceptability consist of 100, (50 students and 50 professional). The scoring scale was: 1 (dislike extremely), 2 (dislike very much), 3 (dislike moderately), 4 (dislike slightly), 5 (neither like or dislike), 6 (liked slightly), 7 (liked moderately), 8 (liked very much), 9 (liked extremely).

Experimental Design and Treatment

Experimental Design

The study used experimental method of research. An experimental method is a method or procedure involving the control or manipulation of conditions for the purpose of studying the relative effects of various treatments applied to the mixture of the different samples (Calderon, 2002).

Experimental Treatment

The study used four (4) treatments with three replications: Treatment A (100% bangus), Treatment B (75% bangus, 25% clam), Treatment C 50% bangus, 50% clams) and Treatment D (25% bangus, 75% clams). The same ingredients and proportion was used in the study except for the proportion of bangus and clam which vary per treatment.

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For the sensory evaluation:

The Bangus-Clam Sisig was prepared in accordance with the protocol for sensory evaluation. Score cards which contain the 5-point scale for the sensory characteristics and nine-point Hedonic scale for the acceptability was used to evaluate the develop product. 30 semi-trained evaluators evaluated the product. Only sensory characteristics such as color, taste, flavor and texture was evaluated.

Data Collection

The data was gathered using score containing the 9-point hedonic scale. It contains the rating of the panelist on color, odor, texture, taste, and general acceptability. The data was gathered after every replication.

Scoring of Variables

The criteria for evaluation as to sensory characteristics of the product was 5-point hedonic scale in terms of color, odor, texture and taste. The rating for general acceptability in 9-point hedonic scale was: 9 for like extremely, 8 for like very much, 7 for like moderately, 6 for like slightly, 5 for neither like nor dislike, 4 for dislike moderately, 3 for dislike slightly, 2 for dislike very much and 1 for dislike extremely (Calmorin, 2006).

The score on each item was interpreted based on the result of the computation as to color, odor, texture, taste and general acceptability using the following scoring interval and verbal interpretation.

Color

Scoring	Verbal Interpretation/Response Categories	Scoring Interval
5	Brown	4.2-5.0
4	Light brown	3.4-4.9
3	Slightly brown	2.6-3.39
2	Moderately brown	1.8-2.59
1	yellow	1.0-1.79

For Odor

Scoring	Verbal interpretation/Response Categories	Scoring Interval
5	No bangus-clam odor	4.2-5.0
4	With slight bangus-clam odor	3.4-4.9
3	With slightly strong bangus-clam odor	2.6-3.39
2	Moderately strong bangus-clam odor	1.8-2.59
1	Extremely strong bangus-clam odor	1.0-1.79

For Texture

Scoring	Verbal Interpretation/Response Categories	Scoring Interval
5	crisp	4.2-5.0
4	Slightly crisp	3.4-4.9
3	soft	2.6-3.39

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2	Fairly hard	1.8-2.59
1	Very hard	1.0-1.79

For Taste

Scoring	Verbal interpretation/Response Categories	Scoring Interval
5	No bangus-clam taste	4.2-5.0
4	With slight bangus-clam taste	3.4-4.9
3	With slightly strong bangus-clam taste	2.6-3.39
2	Moderately strong bangus-clam taste	1.8-2.59
1	Extremely strong bangus-clam taste	1.0-1.79

For Acceptability

Scoring	Score Interval	Verbal interpretation/Response Categories
9	8.12 – 9.0	Like Extremely
8	7.23 – 8.11	Like Very Much
7	6.34 – 7.22	Like Moderately
6	5.45 – 6.33	Like Slightly
5	4.56 – 5.44	Neither Like nor Dislike
4	3.67 – 4.55	Dislike Slightly
3	2.78 – 3.66	Dislike Moderately
2	1.89 – 2.77	Dislike Very Much
1	1.00 – 1.88	Dislike Extremely

Statistical Tools and Analysis

The data was analyzed using weighted mean and Analysis of Variance (ANOVA).

The weighted mean was used to determine the acceptability of the sensory characteristics such as color, odor, taste, texture and general acceptability of the product.

The analysis of variance was used to determine the significant difference among treatment in terms of color, odor, taste, and texture.

For microbial analysis and shelf life:

The sample was sent to DOST or any available accredited agency for analysis. The package was dependent on the requirement set by accredited agency.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use grant chart.

Research Objectives	Year 1 Year 2											
	Mar	Apr	May	Jun	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb

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1. Preparation of proposal and other related activities													
2. Product Development													
3. Sensory Assessment/Organoleptic test													
4. Proximate analysis													
5. Microbial analysis													
6. Preparation & writing of data and results													
7. Presentation of Results and Recommendations													
8. IP Protection Application													

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

To provide healthier options as a response to the trend in the market. The researcher utilizes bangus and clam as the meat for sisig production.

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Process will be taught to the target clientele through extension program of the college, however, prior to the extension activity, process of the product will be applied for the utility model of IPOPHIL.

(17) Target Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

The target beneficiaries of this study are entrepreneurs, communities, and households.

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount(PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1	3,000

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Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
Transportation expense		500.00
Proximate Analysis	8	7,750.00
Microbial Analysis	2	1,100.00
Packaging		1,500.00
TOTAL		13,850.00
CAPITAL OUTLAY		
Bangus	10kls@ 250	2,500.00
Clam	10kls @ 120	1,200.00
Garlic	2kl @ 120	240.00
Chili	1kg@250	250.00
Onion	2kls@150	300.00
Liver Spread	10cans@55	550.00
Soy Sauce	1Gal@250	250.00
Black pepper @McCormick	1kg@1,800	1,800.00
Vinegar	1Gal@180	180.00
Salt	1kl@50	50.00
Annatto oil	1L @ 300	300.00
TOTAL		7,620.00
OVERALL TOTAL		21,470.00

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

PH22018000610U1 A COMPOSITION FOR MUD CLAMS (ANODONTIA EDENTULA) SALAD (SISIG) WITH COCONUT MILK

PH22019001306U1 PROCESS OF PRODUCING MUSHROOM SISIG

PH22016000709U1 FORMULATION OF MILKFISH (CHANOS CHANOS) DUMPLING FILLING

(21) Capsule Curriculum Vitae (One-page brief C.V. only)

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EDUCATIONAL BACKGROUND

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Doctoral Degree	:	Central Philippine University Jaro, Iloilo Doctor of Tourism and Hospitality Management	2020-2021
Master Degree	:	Colegio de la Purisima Conception Roxas City, Capiz Master of Business Administration (M.B.A.)	2010
College	:	Colegio Del Sagrado Corazon de Jesus Iloilo City Bachelor of Science in Hotel and Restaurant Management	1996
Work Experience:			
Hercor College: Faculty			2005-2015
Subject Handle: Asian Cuisine, International Cuisine, Food and Beverage, Catering Management, Costing Management, Bakery Management, Total Quality Management, Bread and Pastry.			
General Manager: Hercor Catering Services			2005-2015
Capiz State University: Professor			2015 to present
Subject Handle: Bakery Management, Food Safety, Principles to Human Nutrition, Entrepreneurship, Trade and Marketing			
Gabriel's Catering and Services-owner			2015 to present
Gabriel's Event Center-owner			
Rafael's Cakes-owner			
Village Coffee-Owner			